

Prathamesh Dinesh Rane

M.Sc. Student, Computational Modeling and Simulation · TU Dresden

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PROFILE

I study Computational Modeling and Simulation at TU Dresden and have worked on aircraft models and flight-simulation components built from black-box flight data. Most of my recent work is in Python: analysing machine learning models, writing graph algorithms and evaluating results over repeated runs. I am looking for a student assistant position or internship where I can work on simulation, data analysis or scientific software alongside my studies.

EDUCATION

Technische Universität Dresden

M.Sc. Computational Modeling and Simulation

Dresden, Germany

expected Mar 2027

- Coursework in machine learning, scientific computing, simulation, data engineering and visualization.

Shah and Anchor Kutchhi Engineering College, University of Mumbai

Bachelor of Engineering in Information Technology

Mumbai, India

2024

- 93% in the C programming exam of the IIT Bombay Spoken Tutorial programme.

EXPERIENCE

Aviation Horizons Limited

Developer

Mumbai, India

Aug 2022 – Aug 2023

- Built aircraft models and flight-simulation components from black-box flight data.
- Cleaned and interpreted recorded flight data and turned it into structured inputs for the simulation.
- Tested and debugged the simulation software together with the rest of the team.

SAKEC Virtual Lab

Creative Co-Lead

Mumbai, India

Nov 2021 – Apr 2022

- Built interactive chemistry simulations and animated experiments for the college's online virtual-lab platform.
- Co-led the creative work and coordinated with the frontend and graphic design teams on maintaining and extending the site.

Freelance web development

Alpha Laboratories, D.D. Vispute College

Mumbai, India

May 2021 – Apr 2022

- Built and maintained two organisation websites (HTML, CSS, JavaScript) based on client requirements.

PROJECTS

Explainability and bias in large language models

Python, Transformers

- Aggregated token-level attention scores across transformer layers and heads.
- Compared neutral and demographic prompt variants to study bias and false refusals, and visualised the results.

Semantic map inference

Python, graph algorithms

- Implemented greedy and randomised graph-inference algorithms under connectivity constraints.
- Ran each method many times and measured how stable the results were using Jaccard similarity and entropy.

Robot manipulation in simulation

Python, RoboCasa, MuJoCo · ongoing

- Setting up simulated kitchen tasks, recording demonstrations and replaying them for robot-learning experiments.

Cutato booking platform

TypeScript, Next.js, PostgreSQL · ongoing

- Full-stack booking system with authentication, scheduling and role-based access.

SKILLS

Programming: Python, C, C++, SQL, TypeScript/JavaScript

Scientific / data: NumPy, Pandas, Matplotlib, Jupyter, graph algorithms, statistical evaluation

ML: transformer models, attention analysis, model evaluation

Simulation: flight simulation, MuJoCo, RoboCasa

Tools: Linux, Git/GitHub, PostgreSQL, VS Code

Languages: English (fluent)